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1. Introduction

The company specialises in approving loans and it aims to maximize profits by approving credit-worthy applicants while minimizing losses from defaults. Loan officers rely on a computer model to assist in their decisions, but with economic conditions shifting, its accuracy is in question. To address this, a new model has been developed. This report analyses the effectiveness of this new computer model by comparing officers using the existing model (control group) with officers using the new model (treatment group). A series of statistical analyses was conducted focusing on key metrics such as Type II errors, F1 Score and Agreement of the officers with the models. This report outlines the methodology, the results obtained and recommendations for future enhancements.

2. Data Preparation

The first step in our analysis was to convert the metric 'Variant' into a factor. This was necessary because 'Variant' represents categorical data (Control vs. Treatment), and statistical functions in R (e.g., t-test) require categorical variables to be explicitly defined as factors.

The next step was to exclude all observations of officers with a zero value in the metric *'fully_complt'* column, as we only focused on officers who had completed reviews for our hypothesis and Overall Evaluation Criteria (OEC). This was important because some loan officers have not completed any reviews, which could have introduced bias into our analysis as the data for these officers were inconsistent. By focusing exclusively on officers with fully completed reviews, we ensured that our comparisons were made within a consistent and comparable group of officers. Thereafter, we performed the aggregation by calculating the normalized values of key metrics (e.g., *'typeII_fin', 'agree_fin'*) by dividing the sum or difference of each metric by the total number of fully completed reviews ('*fully_complt'*). The loan officers were randomly assigned as per the case study and these loan officers represent the experimental unit.

3. Defining the Overall Evaluation Criteria (OEC)

3.1 Minimizing the number of bad loans

The main objective of the company is to reduce losses and increase profits. To achieve this, we focused on minimizing Type II errors (false negatives), which occur when a loan officer approves a loan that later defaults and results in financial losses for the company. Therefore, we established that Type II errors after seeing computer predictions (‘*typeII_fin’*) as one of our OEC. This metric measures the effectiveness of the model in reducing bad loans after loan officers have reviewed the computer's predictions.

In the context of consumer lending, type II errors (approving bad loans) are more detrimental to the company’s financial health than type I errors (rejecting good loans) because:

- A Type II error results in a direct financial loss because the loan is not repaid. In contrast, a Type I error represents a missed opportunity for profit, but no direct loss.
- While rejecting good loans (Type I errors) can frustrate potential customers, approving bad loans (Type II errors) can lead to a higher default rate, which can damage customer trust in the long term.

3.2 Analyzing the F1 Score for Loan Classification Performance

Given that one of the company's goals is to increase the number of good loans while reducing the number of bad loans, a higher F1 score is desirable because it balances both precision (reducing false positives) and recall (reducing false negatives), thereby providing a more comprehensive measure of the model's ability to correctly classify loans.

3.3 Agreement of the officers with the model’s prediction

The confidence of the officers in the model is correlated with how correctly the model categorises the loan and how often the officers agree with the prediction of the model. If the number of agreement of officers with the model increases, the trust and confidence of the officers in the model will also increase, which will also reflect that the officers are having lesser number of conflicts with the model. Therefore, we have considered the metric Agreement (‘*agree_fin* – *agree_init*’) as our third OEC as this represents the change in the agreement of the officers with the model before (initial) and after (final) seeing the model’s prediction.

4. Formulating the Hypothesis

Based on the given problem statement and data, the hypothesis we have formulated are:

1. The new model will decrease each loan officer's Type II errors after seeing computer predictions for those officers who have fully completed loan reviews.

2. The new model is expected to improve the F1 Score for officers who have fully completed loan reviews.

3. Officers with fully completed loan reviews will have greater agreement between their decision and the prediction of the new model (Treatment) compared to the previous model (Control).

Our primary hypothesis supports the company's goal of minimizing losses by reducing Type II errors and ensuring better adaptation to the new model. The second hypothesis focuses on the officers' agreement with both models, while the third monitors changes in the F1 Score, providing valuable insights into whether more good loans are being approved and more bad loans are being rejected.

5. Performing OECs and Data Analysis

5.1 Analyzing Type II Errors

To test our first hypothesis, we performed a Welch's t-test to compare the normalized Type II errors *('typeII_fin_norm')* between the Control (existing model) and Treatment (new model) variants. The results were Welch $t(10.85)=3.52$, $p=0.00489$, with a difference of 0.039 between the means and CI [0.014 – 0.063]. The results indicate that Type II errors were significantly larger in the Control group (mean = 0.124) compared to the Treatment group (mean = 0.0856). This suggests that the new model effectively reduces Type II errors.

To further validate our findings, we performed a pairwise t-test and adjusted the p-values for multiple comparisons using the Bonferroni-Holm (BH) method. The results remained statistically significant, confirming that the Treatment group significantly reduced Type II errors compared to the Control group by 31.2%. Kohavi, Tang, & Xu, (2020) emphasize that statistical significance should be assessed using robust techniques such as sequential testing with always valid p-values.

We also calculated the effect size using Cohen's d (Cohen, 1988), which measures the magnitude of the difference between the two groups. The effect size was d = -1.72, indicating a large and practically meaningful difference between the Control and Treatment groups.

5.1.1 Analyzing the Recall

Since we are focused on analyzing the changes in Type II errors with the new model, we also computed Recall, which represents the proportion of actual bad loans that the loan officers correctly identified. Recall is defined as:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Where:

- True Positives (TP): Bad loans correctly identified (rejected).
- False Negatives (FN): Bad loans incorrectly approved (Type II errors).

We know that in the '*badloans_num'* there are bad loans that were correctly identified (TP) and bad loans that were mistakenly approved (FN), so to calculate the true positives we subtract from '*badloans_num'* the *'typeII_fin'*.

When we performed the t-test for Recall we got a significant result, Welch t(128.48)= - 3.613, p < 0.005, with a difference of 0.1306 between the means, CI [0.059 – 0.2022]. So, there is clearly a significant difference between Control (mean = 0.637) and Treatment (mean = 0.767) groups.

This finding aligns with the improvement in Type II errors that we observed earlier. The Recall results suggest that the model helps loan officers reduce the likelihood of approving bad loans.

5.2 Analyzing F1 Score

To test our second hypothesis, we evaluate the F1 score with the new model. We first calculated Precision, which is defined as:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

Where:

- True Positives (TP): As in Recall, are the bad loans correctly identified (rejected).
- False Positives (FP): Good loans incorrectly rejected (Type I errors).

Next, using the Precision and Recall, we calculated the F1 Score, which is defined as:

$$\text{F1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

After conducting the t-test, we obtained a significant result again, Welch $t(180.5)= -10.119$, $p < 0.005$, with a difference of 0.16277 between the means, CI [0.131 – 0.194].

Additionally, we performed a pairwise t-test comparing the F1 scores between the Treatment and Control groups, which showed a highly significant difference, with a p-value of less than $2e \times 10^{-16}$. This indicates that the Treatment group (mean = 0.646) significantly outperforms the Control group (mean = 0.483) by 33.6%. To account for multiple comparisons, the p-value was adjusted using the BH method.

After calculating the effect size with Cohen's d (Cohen, 1988), we obtained a value of 1.15, suggesting a significant and practically meaningful difference between the Control and Treatment groups.

Therefore, similar to Recall, there is a significant difference in F1 scores of both the variants. The significant result indicates that the model is effectively improving loan decisions, leading to a reduction in bad loans while increasing the approval of good loans. These improvements align with our previous findings, reinforcing that the model is enhancing decision-making accuracy and demonstrating its overall effectiveness in optimizing loan assessments.

5.3 Analyzing Officer's Agreement with the Model

This metric provides a clear picture of the loan officer's trust and agreement with the model's predictions. The results were, Welch t(32.688)= -3.2744 , p=0.0025, with a difference of 0.08 in the means, CI [-0.1292 – -0.0301]. This shows that agreement is significantly higher in the Treatment group (mean = 0.1479) compared to the Control group (mean = 0.068). This indicates that the new model increases agreement, further supporting its effectiveness in building trust among loan officers.

The bar chart (*figure 1.*) shows the difference between the mean values of the officer's agreement with the model and type II errors. Te chart shows that based on the current dataset, treatment variant is performing significantly better than control variant.

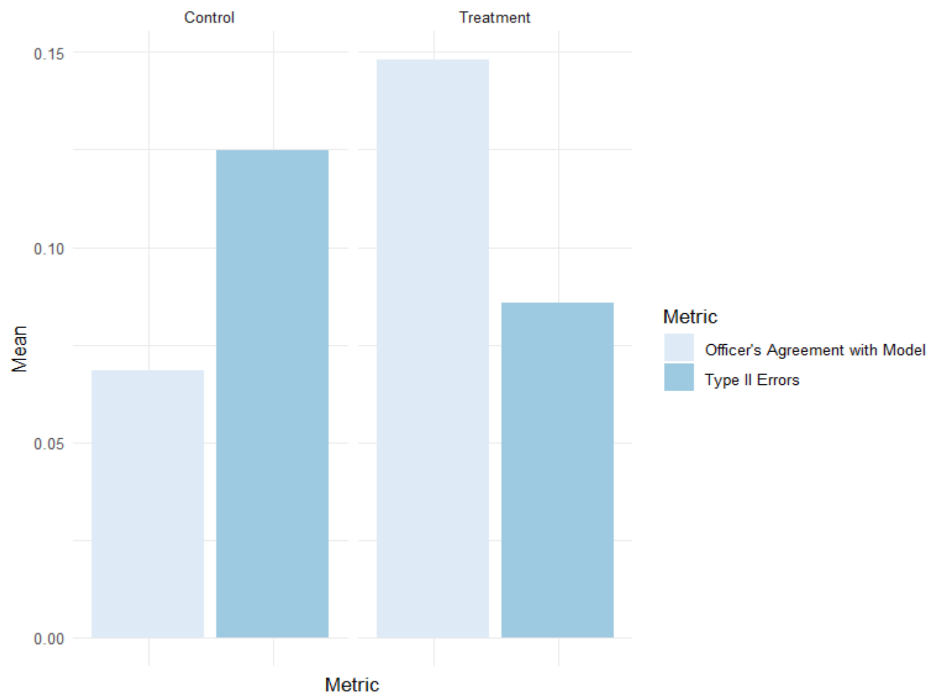

Figure 1. Mean of the normalized metrics by Variant

6. Recommendations

Our analysis shows that the treatment (new) model significantly outperforms the control (older) model. There was a 31.2% reduction in Type II errors, meaning fewer bad loans were mistakenly approved. Additionally, the F1 score increased by 33.6%, reflecting improved precision and recall in loan classification. A high effect size (Cohen's $d > 1$) confirms a substantial performance shift, highlighting the model's meaningful impact. Given these results, the experiment can be stopped as the evidence strongly favours the new model. However, continued monitoring over a longer period would ensure sustained improvements and adaptability to economic changes. Since the data covers only 10 days, it may not capture the full variability in loan applications. Expanding the dataset and incorporating broader economic factors would enhance reliability and provide a more comprehensive assessment of the model's effectiveness.

6.1 Model Enhancements

- **Incorporating Additional Metrics:** The current experimental design does not fully account for key financial metrics, such as the profit impact of loan decisions. Future experiments should integrate metrics that directly connect decision quality to financial performance.

- **Improving Model Training:** Enhancing the model's performance requires incorporating a larger dataset or leveraging advanced algorithms to improve its ability to accurately identify high-risk loans. This can significantly improve the quality of predictions of the model.

- **Randomization:** It is crucial to maintain proper randomization in assigning loan officers to the control and treatment groups. Randomization helps eliminate biases that could arise from factors such as experience level, workload, or individual decision-making tendencies, ensuring that any observed differences in performance can be attributed to the new model rather than external variables. One way to achieve this is by using stratified randomization, where loan officers are categorized based on key characteristics (e.g., years of experience or historical error rates) before being randomly assigned to a group.

7. Conclusion

The A/B testing confirms that the new computer model significantly improves loan approval decisions, reducing financial risk (Type II errors) and provides a good balance between precision and recall as indicated by a higher F1 score. Loan officers also showed greater alignment with new model's predictions, indicating increased confidence in its recommendations. Given these results, the experiment can be stopped, and the model should be adopted; however, with continued monitoring to ensure adaptability to economic changes. Further refinements, such as integrating financial metrics, using stratified randomization and optimizing model training, will enhance decision-making and strengthen the company's lending strategy.

References

- **Tang, D., Kohavi, R. & Xu, Y.** (2020) *Trustworthy online controlled experiments: A practical guide to A/B testing*. Cambridge University Press.

- **Cohen, J.** (1988) *Statistical power analysis for the behavioral sciences*. 2nd edn. Hillsdale, NJ: Lawrence Erlbaum Associates.

Appendix

```r
library(tidyverse)

library(dplyr)
library(effectsize)

library(ggplot2)
library(pwr)
```

Step 1. Data Prep

```r
exp <- read_csv("ADAproject_-5_data.csv")

## Rows: 470 Columns: 22
## ── Column specification ────────────────────────────────────────────────
────────
## Delimiter: ","
## chr  (2): Variant, loanofficer_id
## dbl (20): day, typeI_init, typeI_fin, typeII_init, typeII_fin, agree_ini
t, a...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.

# View the type of data of the different columns
str(exp)

## spc_tbl_ [470 × 22] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Variant              : chr [1:470] "Treatment" "Treatment" "Treatment
" "Treatment" ...
##  $ loanofficer_id       : chr [1:470] "qamcqdoe" "qamcqdoe" "qamcqdoe" "
qamcqdoe" ...
##  $ day                  : num [1:470] 1 2 3 4 5 6 7 8 9 10 ...
##  $ typeI_init           : num [1:470] 0 2 3 1 0 0 0 0 0 0 ...
##  $ typeI_fin            : num [1:470] 0 2 3 2 2 1 1 3 1 0 ...
##  $ typeII_init          : num [1:470] 2 3 0 1 0 4 1 4 4 2 ...
##  $ typeII_fin           : num [1:470] 2 3 0 1 0 0 0 1 1 1 ...
##  $ agree_init           : num [1:470] 7 8 9 8 8 5 8 4 6 9 ...
##  $ agree_fin            : num [1:470] 10 8 9 9 10 10 10 10 10 10 ...
##  $ conflict_init        : num [1:470] 2 2 1 2 2 5 2 6 4 1 ...
##  $ conflict_fin         : num [1:470] 0 2 1 1 0 0 0 0 0 0 ...
##  $ revised_per_ai       : num [1:470] 2 0 0 1 2 5 2 6 4 1 ...
##  $ revised_agst_ai      : num [1:470] 0 0 0 0 0 0 0 0 0 0 ...
##  $ fully_complt         : num [1:470] 9 10 10 10 10 10 10 10 10 10 ...
##  $ confidence_init_total: num [1:470] 706 911 710 694 683 743 993 1000 1
000 1000 ...
##  $ confidence_fin_total : num [1:470] 913 974 970 961 1000 1000 1000 100
0 1000 1000 ...
##  $ complt_init          : num [1:470] 9 10 10 10 10 10 10 10 10 10 ...
##  $ complt_fin           : num [1:470] 10 10 10 10 10 10 10 10 10 10 ...
##  $ ai_typeI             : num [1:470] 0 1 2 1 2 1 1 3 1 0 ...
```

```
##  $ ai_typeII            : num [1:470] 2 2 0 1 0 0 0 1 1 1 ...
##  $ badloans_num         : num [1:470] 4 5 2 3 0 4 1 4 4 3 ...
##  $ goodloans_num        : num [1:470] 6 5 8 7 10 6 9 6 6 7 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   Variant = col_character(),
##   ..   loanofficer_id = col_character(),
##   ..   day = col_double(),
##   ..   typeI_init = col_double(),
##   ..   typeI_fin = col_double(),
##   ..   typeII_init = col_double(),
##   ..   typeII_fin = col_double(),
##   ..   agree_init = col_double(),
##   ..   agree_fin = col_double(),
##   ..   conflict_init = col_double(),
##   ..   conflict_fin = col_double(),
##   ..   revised_per_ai = col_double(),
##   ..   revised_agst_ai = col_double(),
##   ..   fully_complt = col_double(),
##   ..   confidence_init_total = col_double(),
##   ..   confidence_fin_total = col_double(),
##   ..   complt_init = col_double(),
##   ..   complt_fin = col_double(),
##   ..   ai_typeI = col_double(),
##   ..   ai_typeII = col_double(),
##   ..   badloans_num = col_double(),
##   ..   goodloans_num = col_double()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```r
## Set categorical variable
exp$Variant <- factor(exp$Variant)


# Check for missing values in the data
(missing_values_exp <- exp %>%
  summarise(across(everything(), ~ sum(is.na(.)))))
```

```
## # A tibble: 1 × 22
##   Variant loanofficer_id   day typeI_init typeI_fin typeII_init typeII_f
in
##     <int>          <int> <int>      <int>     <int>       <int>      <in
t>
## 1       0              0     0          0         0           0
0
## # ℹ 15 more variables: agree_init <int>, agree_fin <int>, conflict_init <
int>,
## #   conflict_fin <int>, revised_per_ai <int>, revised_agst_ai <int>,
## #   fully_complt <int>, confidence_init_total <int>,
## #   confidence_fin_total <int>, complt_init <int>, complt_fin <int>,
## #   ai_typeI <int>, ai_typeII <int>, badloans_num <int>, goodloans_num <
int>
```

```r
# There is no missing value in the data

#Eliminate all rows that have a 0 value in the fully_complt column

cleaned_data <- exp %>%
  filter(fully_complt > 0)

summary(cleaned_data)
```

```
##       Variant    loanofficer_id          day         typeI_init
##  Control   :100   Length:380         Min.   : 1.0   Min.   : 0.000
##  Treatment:280   Class :character   1st Qu.: 3.0   1st Qu.: 1.000
##                  Mode  :character   Median : 5.5   Median : 2.000
##                                     Mean   : 5.5   Mean   : 2.613
##                                     3rd Qu.: 8.0   3rd Qu.: 4.000
##                                     Max.   :10.0   Max.   :10.000
##    typeI_fin      typeII_init      typeII_fin       agree_init
##  Min.   :0.000   Min.   :0.000   Min.   :0.0000   Min.   : 0.000
##  1st Qu.:1.000   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.: 6.000
##  Median :2.000   Median :1.000   Median :1.0000   Median : 8.000
##  Mean   :2.355   Mean   :1.008   Mean   :0.9026   Mean   : 6.976
##  3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:1.0000   3rd Qu.: 9.000
##  Max.   :8.000   Max.   :4.000   Max.   :3.0000   Max.   :10.000
##    agree_fin       conflict_init    conflict_fin  revised_per_ai
##  Min.   : 2.000   Min.   :0.000   Min.   :0.00   Min.   :0.000
##  1st Qu.: 7.000   1st Qu.:1.000   1st Qu.:0.75   1st Qu.:0.000
##  Median : 9.000   Median :2.000   Median :1.00   Median :1.000
##  Mean   : 8.166   Mean   :2.397   Mean   :1.55   Mean   :1.008
##  3rd Qu.: 9.000   3rd Qu.:3.000   3rd Qu.:2.00   3rd Qu.:2.000
##  Max.   :10.000   Max.   :8.000   Max.   :8.00   Max.   :8.000
##  revised_agst_ai   fully_complt    confidence_init_total confidence_fin_
total
##  Min.   :0.0000   Min.   : 1.000   Min.   :  50.0        Min.   :  80.0
##  1st Qu.:0.0000   1st Qu.:10.000   1st Qu.: 447.5        1st Qu.: 581.8
##  Median :0.0000   Median :10.000   Median : 635.5        Median : 713.5
##  Mean   :0.1053   Mean   : 9.374   Mean   : 613.8        Mean   : 695.0
##  3rd Qu.:0.0000   3rd Qu.:10.000   3rd Qu.: 770.2        3rd Qu.: 853.2
##  Max.   :4.0000   Max.   :10.000   Max.   :1000.0        Max.   :1000.0
##   compl t_init        complt_fin         ai_typeI        ai_typeII
##  Min.   : 1.000   Min.   : 2.000   Min.   :0.000   Min.   :0.000
##  1st Qu.:10.000   1st Qu.:10.000   1st Qu.:1.000   1st Qu.:0.000
##  Median :10.000   Median :10.000   Median :1.000   Median :1.000
##  Mean   : 9.405   Mean   : 9.716   Mean   :1.516   Mean   :1.011
##  3rd Qu.:10.000   3rd Qu.:10.000   3rd Qu.:2.000   3rd Qu.:2.000
##  Max.   :10.000   Max.   :10.000   Max.   :5.000   Max.   :3.000
##   badloans_num goodloans_num
##  Min.   :0     Min.   : 5
##  1st Qu.:2     1st Qu.: 6
##  Median :3     Median : 7
##  Mean   :3     Mean   : 7
##  3rd Qu.:4     3rd Qu.: 8
##  Max.   :5     Max.   :10
```

```
## Aggregate the data to Officer Level by Type II error
officer_level_data <- cleaned_data %>%
  group_by(Variant, loanofficer_id) %>%
  summarise(
    typeII_fin_sum = sum(typeII_fin, na.rm = TRUE),
    fully_complt_sum = sum(fully_complt, na.rm = TRUE)
  ) %>%
  mutate(typeII_fin_norm = typeII_fin_sum / fully_complt_sum) %>%
  filter(!is.na(typeII_fin_norm))

## `summarise()` has grouped output by 'Variant'. You can override using th
e
## `.groups` argument.

# Perform a t-test

t.test(typeII_fin_norm ~ Variant,
       data = officer_level_data,
       var.equal = FALSE) # assuming samples have unequal variances

##
##  Welch Two Sample t-test
##
## data:  typeII_fin_norm by Variant
## t = 3.5206, df = 10.849, p-value = 0.004893
## alternative hypothesis: true difference in means between group Control a
nd group Treatment is not equal to 0
## 95 percent confidence interval:
##  0.01456457 0.06336915
## sample estimates:
##   mean in group Control mean in group Treatment
##              0.12472870              0.08576184

#There is a significant difference between control and treatment. TypeII_fi
n error are reduced with the new model.


# Perform pairwise t-tests with adjustment for multiple comparisons
pairwise_ttest <- pairwise.t.test(
x = officer_level_data$typeII_fin_norm,
g = officer_level_data$Variant,
var.equal = FALSE,
p.adjust.method = "BH" # Options: "holm", "bonferroni", "BH", etc.
)
# Display results
print(pairwise_ttest)

##
##  Pairwise comparisons using t tests with pooled SD
##
## data:  officer_level_data$typeII_fin_norm and officer_level_data$Variant
##
##           Control
## Treatment 4e-05
```

```
##
## P value adjustment method: BH
```

```r
#There is a statistically significant difference between the groups in term
s of typeII_fin_norm after adjusting for multiple comparisons using the BH
method. The treatment appears to have a strong effect on typeII_fin_norm co
mpared to the control.


# Compute mean OEC for each Variant
mean_OEC_each_Variant <- officer_level_data %>%
group_by(Variant) %>%
summarise(mean_typeII_fin = mean(typeII_fin_norm))
# View mean OEC
print(mean_OEC_each_Variant)
```

```
## # A tibble: 2 × 2
##   Variant   mean_typeII_fin
##   <fct>               <dbl>
## 1 Control             0.125
## 2 Treatment           0.0858
```

```r
# Compute pairwise % differences in OEC
pairwise_diff <- mean_OEC_each_Variant %>%
summarise(
Diff_T1_Control = mean_typeII_fin[Variant == "Treatment"] - mean_typeII_fin
[
Variant == "Control"],
Perc_T1_Control = (Diff_T1_Control / mean_typeII_fin[Variant == "Control"])
* 100,
)
# View pairwise differences
print(pairwise_diff)
```

```
## # A tibble: 1 × 2
##   Diff_T1_Control Perc_T1_Control
##             <dbl>           <dbl>
## 1         -0.0390           -31.2
```

```r
#Treatment (new model) significantly reduced (p < 0.01) typeII_fin error co
mpared to Control (existing model) by 31.2%.


#Calculate the effect size
Control = officer_level_data$typeII_fin_norm[officer_level_data$Variant ==
"Control"]
T1 = officer_level_data$typeII_fin_norm[officer_level_data$Variant == "Trea
tment"]
cohens_d(T1, Control) # compute effect size of difference between Treatment
& Control
```

```
## Cohen's d |         95% CI
## -------------------------
## -1.72     | [-2.54, -0.89]
```

```
##
## - Estimated using pooled SD.

#The large effect size (Cohen's d = -1.72) suggests that the difference bet
ween the groups is not only statistically significant but also practically
meaningful.

## Aggregate the data to Officer level by Type I error

officer_level_data2<- cleaned_data %>%
  group_by(Variant, loanofficer_id) %>%
  summarise(
    typeI_fin_sum = sum(typeI_fin, na.rm = TRUE),
    fully_complt_sum = sum(fully_complt, na.rm = TRUE)
  ) %>%
  mutate(typeI_fin_norm = typeI_fin_sum / fully_complt_sum) %>%
  filter(!is.na(typeI_fin_norm))

## `summarise()` has grouped output by 'Variant'. You can override using th
e
## `.groups` argument.

# Perform a t-test

t.test(typeI_fin_norm ~ Variant,
       data = officer_level_data2,
       var.equal = FALSE) # assuming samples have unequal variances

##
##  Welch Two Sample t-test
##
## data:  typeI_fin_norm by Variant
## t = 3.7781, df = 10.84, p-value = 0.003136
## alternative hypothesis: true difference in means between group Control a
nd group Treatment is not equal to 0
## 95 percent confidence interval:
##  0.06976741 0.26534477
## sample estimates:
##   mean in group Control mean in group Treatment
##               0.3757783               0.2082222

#There is a significant difference between control and treatment. TypeI_fin
errors are reduced with the new model.

# Aggregate the data to Officer level by net agreement with computer predic
tions

officer_level_data3 <- cleaned_data %>%
  group_by(Variant, loanofficer_id) %>%
  summarise(
    agree_fin_sum = sum(agree_fin, na.rm = TRUE),  # Summation of agreement
after seeing computer predictions
    agree_init_sum = sum(agree_init, na.rm = TRUE), # Summation of agreemen
t before seeing computer predictions
```

```
    fully_complt_sum = sum(fully_complt, na.rm = TRUE)
  ) %>%
  mutate(
    Net_agreement = ((agree_init_sum - agree_fin_sum)/fully_complt_sum),   #
Net Agreement in relation to the full applications reviewed.
  ) %>%
  filter(!is.na(Net_agreement) )  # Remove rows with NA values

## `summarise()` has grouped output by 'Variant'. You can override using th
e
## `.groups` argument.

# Perform the t-test for Net_agreement
t.test(Net_agreement ~ Variant,
                             data = officer_level_data3,
                             var.equal = FALSE)

##
##  Welch Two Sample t-test
##
## data:  Net_agreement by Variant
## t = 3.2744, df = 32.688, p-value = 0.002507
## alternative hypothesis: true difference in means between group Control a
nd group Treatment is not equal to 0
## 95 percent confidence interval:
##  0.03015947 0.12923010
## sample estimates:
##   mean in group Control mean in group Treatment
##             -0.06829281             -0.14798759

#There is a significant increase in the net agreement between treatment and
control.
#Higher agreement could indicate better alignment with the model's guidance
.

# Calculate the Recall

officer_level_data7 <- cleaned_data %>%
  group_by(Variant, loanofficer_id, day) %>%
  summarise(
    typeII_fin_sum = sum(typeII_fin, na.rm = TRUE),  # False Negatives (FN)
    badloans_num_sum = sum(badloans_num, na.rm = TRUE)  # Total bad loans
  ) %>%
  mutate(
    TP = badloans_num_sum - typeII_fin_sum,  # True Positives
    Recall = TP / badloans_num_sum  # Recall = TP / (TP + FN)
  ) %>%
  filter(!is.na(Recall))  # Remove rows with NA recall values

## `summarise()` has grouped output by 'Variant', 'loanofficer_id'. You can
## override using the `.groups` argument.
```

```r
t.test(Recall ~ Variant,
       data = officer_level_data7,
       var.equal = FALSE)

##
##  Welch Two Sample t-test
##
## data:  Recall by Variant
## t = -3.613, df = 128.48, p-value = 0.0004329
## alternative hypothesis: true difference in means between group Control a
nd group Treatment is not equal to 0
## 95 percent confidence interval:
##  -0.20220688 -0.05910529
## sample estimates:
##   mean in group Control mean in group Treatment
##               0.6373333               0.7679894

#There is a significant difference in recall between the control and treatm
ent groups (p < 0.005). This suggests that the new model significantly impr
ove the ability to identify bad loans.

# Calculate the Precision

officer_level_data8 <- cleaned_data %>%
  group_by(Variant, loanofficer_id, day) %>%
  summarise(
    typeI_fin_sum = sum(typeI_fin, na.rm = TRUE),  # False Positives (FP)
    typeII_fin_sum = sum(typeII_fin, na.rm = TRUE), # False Negatives (FN)
    badloans_num_sum = sum(badloans_num, na.rm = TRUE)  # Total bad loans
  ) %>%
  mutate(
    TP = badloans_num_sum - typeII_fin_sum,  # True Positives
    Precision = TP / (TP + typeI_fin_sum)  # Precision = TP / (TP + FP)
  ) %>%
  filter(!is.na(Precision))  # Remove rows with NA recall values

## `summarise()` has grouped output by 'Variant', 'loanofficer_id'. You can
## override using the `.groups` argument.

t.test(Precision ~ Variant,
       data = officer_level_data8,
       var.equal = FALSE)

##
##  Welch Two Sample t-test
##
## data:  Precision by Variant
## t = -8.9712, df = 292.5, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Control a
nd group Treatment is not equal to 0
## 95 percent confidence interval:
##  -0.2530790 -0.1620151
## sample estimates:
```

```
##   mean in group Control mean in group Treatment
##               0.3425635               0.5501105

# F1 Score:

f1_score_data <- officer_level_data7 %>%
  select(Variant, loanofficer_id, day, Recall) %>%
  inner_join(officer_level_data8 %>% select(Variant, loanofficer_id, day, P
recision),
             by = c("Variant", "loanofficer_id", "day")) %>%
  mutate(
    F1_Score = ifelse((Precision + Recall) == 0, NA, 2 * (Precision * Recal
l) / (Precision + Recall))  # Avoid division by zero
  ) %>%
  filter(!is.na(F1_Score))  # Remove rows with NA F1 values

t.test(F1_Score ~ Variant,
       data = f1_score_data,
       var.equal = FALSE)

##
##  Welch Two Sample t-test
##
## data:  F1_Score by Variant
## t = -10.119, df = 180.52, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Control a
nd group Treatment is not equal to 0
## 95 percent confidence interval:
##  -0.1945216 -0.1310375
## sample estimates:
##   mean in group Control mean in group Treatment
##               0.4838058               0.6465854

#F1 score and Precision are also significant, as Recall.


# Perform pairwise t-tests with adjustment for multiple comparisons
pairwise_ttest2 <- pairwise.t.test(
x = f1_score_data$F1_Score,
g = f1_score_data$Variant,
var.equal = FALSE,
p.adjust.method = "BH" # Options: "holm", "bonferroni", "BH", etc.
)
# Display results
print(pairwise_ttest2)

##
##  Pairwise comparisons using t tests with pooled SD
##
## data:  f1_score_data$F1_Score and f1_score_data$Variant
##
##           Control
## Treatment <2e-16
```

```
##
## P value adjustment method: BH

# After doing the pairwise t-test for multiple comparisons adjusting for th
e Bonferroni-Holm method the difference between control and treatment still
remains significant.

# Compute mean OEC for each Variant
mean_OEC_each_Variant2 <- f1_score_data %>%
group_by(Variant) %>%
summarise(mean_F1_Score = mean(F1_Score))
# View mean OEC
print(mean_OEC_each_Variant2)

## # A tibble: 2 × 2
##   Variant   mean_F1_Score
##   <fct>             <dbl>
## 1 Control           0.484
## 2 Treatment         0.647

# Compute pairwise % differences in OEC
pairwise_diff2 <- mean_OEC_each_Variant2 %>%
summarise(
Diff_T1_Control = mean_F1_Score[Variant == "Treatment"] - mean_F1_Score[
Variant == "Control"],
Perc_T1_Control = (Diff_T1_Control / mean_F1_Score[Variant == "Control"]) *
100,
)
# View pairwise differences
print(pairwise_diff2)

## # A tibble: 1 × 2
##   Diff_T1_Control Perc_T1_Control
##             <dbl>           <dbl>
## 1           0.163            33.6

#Treatment (new model) significantly improve (p < 0.01) F1 Score compared t
o Control (existing model) by 33,6%.

#Calculate the effect size
Control = f1_score_data$F1_Score[f1_score_data$Variant == "Control"]
T1 = f1_score_data$F1_Score[f1_score_data$Variant == "Treatment"]
cohens_d(T1, Control) # compute effect size of difference between Treatment
& Control

## Cohen's d |       95% CI
## ------------------------
## 1.15      | [0.89, 1.41]
##
## - Estimated using pooled SD.

#The large effect size (Cohen's d = 1.15) suggests that the difference betw
een the groups is not only statistically significant but also practically m
eaningful.
```

```
# Plotting the means of Type II error, Type I error and Net Agreement

plot <- cleaned_data %>%
  group_by(Variant, loanofficer_id) %>%
  summarise(
    # Calculate the sums of the relevant variables
    typeII_fin_sum = sum(typeII_fin, na.rm = TRUE),
    typeI_fin_sum = sum(typeI_fin, na.rm = TRUE),
    agree_fin_sum = sum(agree_fin, na.rm = TRUE),
    agree_init_sum = sum(agree_init, na.rm = TRUE),
    fully_complt_sum = sum(fully_complt, na.rm = TRUE)
  ) %>%
  # Filter out rows where the sum of 'fully_complt' is zero to avoid divisi
on by zero
  filter(fully_complt_sum != 0) %>%
  # Normalize the metrics
  mutate(
    typeII_fin_norm = typeII_fin_sum / fully_complt_sum,
    typeI_fin_norm = typeI_fin_sum / fully_complt_sum,
    Net_agree_norm = (agree_fin_sum - agree_init_sum) / fully_complt_sum
  ) %>%
  ungroup()  # Ungroup to avoid accidental grouping issues

## `summarise()` has grouped output by 'Variant'. You can override using th
e
## `.groups` argument.

# Calculate the mean of the normalized metrics for each Variant (Control vs
Treatment)
aggregated_data <- plot %>%
  group_by(Variant) %>%
  summarise(
    typeII_fin_mean = mean(typeII_fin_norm, na.rm = TRUE),
    Net_agree_mean = mean(Net_agree_norm, na.rm = TRUE)
  ) %>%
  # Reshape the data into long format for easy plotting
  pivot_longer(
    cols = c(starts_with("type"), starts_with("Net")),
    names_to = "Metric",
    values_to = "Mean_Value"
  )

# Convert Metric column to a factor for proper labeling
aggregated_data$Metric <- factor(aggregated_data$Metric,
                                 levels = c("Net_agree_mean", "typeII_fin_m
ean"),
                                 labels = c("Officer's agreement with Model
", "Type II Errors"))

# Plot the means for each metric, separated by Variant (Control vs Treatmen
t)
ggplot(aggregated_data, aes(x = Metric, y = Mean_Value, fill = Metric)) +
  geom_bar(stat = "identity", position = "dodge") +
```

```r
  facet_wrap(~ Variant) +  # Facet by Variant (Control vs Treatment)
  labs(
    title = "Mean of Normalized Metrics by Variant",
    x = "Metric",
    y = "Mean",
    fill = "Metric"
  ) +
  scale_fill_brewer(palette = "Blues") +  # Use 'Blues' color palette
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),  # Remove x-axis labels (metric names)
  )
```